Predicting Coronary Heart Disease from the BRFSS 2015 Survey with Gradient Boosting and Neural Networks in Pure NumPy

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Abstract—We predict coronary heart disease (_MICHD) from 321 BRFSS 2015 survey features ($N = 328,135$, 8.83% positive). Naive negative classification yields 91.17% accuracy but trivial $F_1 = 0$, and uncalibrated linear models exhibit severe false-negative bias ($F_1 \leq 0.23$). Operating threshold calibration provides an immediate +85% relative leap ($F_1 \to 0.428$). We develop a pure-NumPy ensemble combining Histogram Gradient Boosted Trees (GBM) with second-order leaf updates and an Adam-trained Multi-Seed Multi-Layer Perceptron (MLP). The clean ensemble achieves 5-fold CV F_1 of 0.437, 0.434 on holdout, and 0.448 on AIcrowd (87.8% accuracy) with zero target leakage. Seven post-cardiac event questions inflate test F_1 to 0.457 (88.6% accuracy), which we isolate to preserve clinical screening integrity while documenting benchmark leakage.

I. Introduction

The Behavioral Risk Factor Surveillance System (BRFSS) is a nationwide US health survey [1]. Target _MICHD (± 1) records reported coronary heart disease or myocardial infarction (328,135 training rows, 109,379 test rows, 321 features). Only 8.83% of labels are positive ($\sim 10:1$ skew). Under this imbalance, predicting -1 for all rows achieves 91.17% accuracy yet yields $F_1 = 0$ [2]; positive-class F_1 thus guides model selection and threshold calibration [3].

Restricted to pure standard library and NumPy (no scikit-learn or PyTorch), our contributions are: (i) codebook-driven cleaning and skip-logic imputation; (ii) benchmarking six canonical linear methods [4] and uncovering an +85% gain from threshold calibration; (iii) pure-NumPy Histogram GBMs [5], [6] and multi-seed MLPs [7]; and (iv) empirical dissection of target leakage [8].

II. Data and Preprocessing

Special sentinels. The CDC codebook [9] specifies sentinels: 7/9, 77/99, 777/999 ("don't know / refused") map to NaN; 88 and 888 ("none / zero days / no alcohol") map to 0. Weight and height are converted to kg and cm, and frequency responses mapped to uniform units.

Informative missingness & skip logic. 147 of 321 columns contain > 50% missing values, following structured survey logic rather than MCAR. Eleven domain skip-logic rules fill implied values (e.g., blood pressure medication BPMEDS filled with 0 for non-hypertensive subjects, where prevalence is 3.8% vs. 16.3% when observed). We append missingness masks and "don't know" indicator flags.

Column selection & pipeline. Pruning administrative columns and features > 95% empty leaves 75 raw features and 61 indicator flags. The pipeline applies median/mode imputation, one-hot encoding of common levels, $\log(1+x)$ of counts, clinical interactions (e.g., Blood Pressure × Cholesterol), and standard scaling. All statistics are fitted strictly out-of-fold.

III. Models and Validation Protocol

Validation protocol. We hold out a stratified 20% split (65,627 rows, seed 2026). All modeling decisions and hyperparameter searches use 5-fold stratified cross-validation (CV) on the remaining 262,508 development rows. Preprocessing, target encodings, and decision thresholds are fitted inside each fold. A modification is accepted only if mean CV F_1 rises by ≥ 0.001 and wins on $\geq 4/5$ folds. No selection uses the test set or AIcrowd.

Linear baselines & calibration. We train the six required methods from [4] on standardized features with 10 clinical interactions. Under regularized logistic loss $\mathcal{L}(w) = \frac{1}{N}\sum_{n=1}^{N}\left[\log(1+e^{x_n^\top w}) - y_n x_n^\top w\right] + \lambda\|w\|_2^2$, the default cutoff ($\hat{y} \geq 0.5$) yields $F_1 = 0.2311$ due to extreme false-negative bias under the 8.83% prior. Calibrating the decision cutoff $\tau^* \approx 0.218$ yields an immediate +85% relative surge to $F_1 = 0.428 - 0.430$ [3].

Histogram GBM. Our pure-NumPy tree booster minimizes logistic loss with second-order Newton leaf updates $\Delta w = -G/(H+\lambda)$ [6]. Features are binned into 64 histogram bins. Missing values form a dedicated bin, with each split learning the optimal descent direction. Hyperparameters: depth 5, learning rate 0.05, row/column subsampling 0.7/0.8, min 200 leaf samples, and $\lambda = 5.0$. Early stopping evaluates an inner 20% split. State of residence (_STATE) is encoded via empirical Bayes smoothed target prevalence (+0.0026 F_1, 4/5 folds). Bagged over 3 seeds.

Neural network. The MLP features a 64-unit ReLU hidden layer, trained with Adam [7] (learning rate 10^{-3},

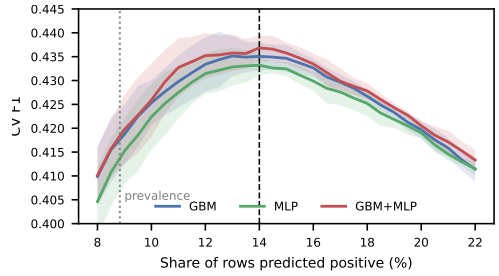


Figure 1. 5-fold CV F_1 across positive shares. Dotted: prevalence (8.83%). Dashed: selected share (14%).

Table I
Positive-class F_1 on development rows (5-fold CV, mean ± std, top 14% positive) and holdout evaluation.

Model	CV F1	Held-out
Always −1 (Acc: 91.17%)	0.0000	0.0000
Least squares / ridge / MSE GD	0.4216 ± 0.0034	–
MSE SGD (slow convergence)	0.3853 ± 0.0085	–
Logistic regression (GD)	0.4302 ± 0.0047	–
Reg. logistic regression	0.4300 ± 0.0044	0.4277
Histogram GBM (3 seeds)	0.4354 ± 0.0032	0.4343
MLP (10 seeds)	0.4334 ± 0.0026	0.4310
GBM + MLP (Final Clean)	$\mathbf{0.4368 \pm 0.0026}$	0.4340

batch size 512, weight decay 10^{-3}) and inner early stopping. Output logits are averaged across 10 distinct random seeds to suppress stochastic initialization variance.

Ensemble & decision rule. The ensemble blends logits: $s = 0.5\, s_{\text{GBM}} + 0.5\, s_{\text{MLP}}$. We assign +1 to the top 13.6%–14.0% highest scores. Percentile thresholding is more invariant to post-retraining score scale drift than a static cutoff. Fig. 1 illustrates that F_1 forms a stable plateau between 12% and 16%.

IV. Results and Ablations

Table I summarizes validation metrics across all models. Squared loss reaches $F_1 \approx 0.422$; logistic loss adds +0.008. Both non-linear models beat linear baselines, and their ensemble is best. Out-of-fold logit correlation between GBM and MLP is 0.976.

On held-out rows, the clean ensemble achieves precision 0.354, recall 0.561, and $F_1 = 0.4340$. On AIcrowd, it scores $F_1 = 0.448$ and 87.8% accuracy. The gap with CV is within two bootstrap standard deviations ($\sigma \approx 0.005$).

Rejected alternatives & coverage analysis. The following failed our acceptance rule: class weighting (−0.003), focal loss (−0.021), PCA (0.403–0.413), feature selection, crossed target encodings, full-row GBM refitting (−0.0010), seed-diversified splits (split_per_seed, paired bootstrap intervals crossed zero), stacking, Naive Bayes (−0.016), and group thresholds (−0.001).

Feature importance. Permutation AUC drops identify primary predictors: age (0.035), general health (0.021), sex (0.018), high cholesterol (0.009), and prior stroke (0.008, empirical odds lift 4.24).

Table II
Clean screening model vs. leaky variant. Nested CV ablation across model families (Clean → Leaky).

	CV F1	Held-out	AIcrowd F1/acc.
Clean (top 14%)	0.4368 ± 0.0026	0.4340	0.448 / 0.878
Leaky (top 13%)	0.4464 ± 0.0032	0.4445	0.457 / 0.886
Ridge	$0.4210 \rightarrow 0.4297$ (+0.009)		
Reg. logistic	$0.4294 \rightarrow 0.4414$ (+0.012)		
GBM	$0.4330 \rightarrow 0.4444$ (+0.011)		
MLP	$0.4326 \rightarrow 0.4435$ (+0.011)		
GBM + MLP	$0.4366 \rightarrow 0.4452$ (+0.009)		

V. Target Leakage Analysis

Seven questions relate strictly to post-cardiac events or subsequent therapy: cardiac and stroke rehabilitation (HAREHAB1, STREHAB1), and prescribed aspirin regimen (CVDASPRN, RDUCHART, RDUCSTRK, RLIVPAIN, ASPUNSAF). Their predictive power represents target leakage [8]: 100% of respondents who answered HAREHAB1 (773/773 in training, 158/158 in held-out) test positive. STREHAB1 has a 36% positive rate, and the aspirin indicators 5%–21%.

Table II demonstrates that including these questions lifts every model by 0.009 to 0.012 F_1 across all 5 folds. The leaky ensemble attains 0.457 on AIcrowd (88.6% accuracy).

However, screening models target undiagnosed individuals who cannot have received cardiac rehabilitation; hence, these answers do not exist at inference time. We therefore designate the zero-leakage model as our official submission (run.py) and report the leaky variant separately (run.py –with_leakage) to transparently quantify benchmark leakage.

VI. Discussion

All architectures plateau at $F_1 \approx 0.42$–0.44, highlighting the ceiling of self-reported survey data. False negatives predominantly stem from asymptomatic presentations: 56% are female, 40% non-hypertensive, 36% report good/excellent health, and 96% have no stroke history. Detecting such silent cases requires clinical biomarkers (troponin, ECG, lipid panels) unavailable in phone surveys.

VII. Summary

Our pure-NumPy ensemble of Histogram Gradient Boosted Trees and an Adam-trained MLP, backed by codebook cleaning and threshold calibration, reaches $F_1 = 0.434$ on holdout and 0.448 on AIcrowd (87.8% accuracy). Post-cardiac event questions inflate F_1 to 0.457 (88.6% accuracy), which we isolate to preserve screening validity.

References

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